

Rileigh Bandy

Boulder, CO 80305

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Education

Ph.D., Computer Science

2024

THE UNIVERSITY OF COLORADO, BOULDER

- Advisor: Rebecca Morrison, Ph.D.
- Dissertation: *Uncertainty Representations in White- and Black-Box Models: Quantifying Model-Form and Measurement Errors in Computational Science*

M.S., Computer Science

2022

THE UNIVERSITY OF COLORADO, BOULDER

3.97/4.0 GPA

B.S., Computer Science

2019

THE UNIVERSITY OF TEXAS AT AUSTIN

3.67/4.0 GPA

Research and Industry Experience

BAE Systems, Inc.

Broomfield, CO

SENIOR SYSTEMS ENGINEER - SPACE & MISSION SYSTEMS

Apr. 2026 - Present

Sandia National Laboratory

Albuquerque, NM

POSTDOCTORAL APPOINTEE

Jun. 2024 - Apr. 2026

RESEARCH AND DEVELOPMENT INTERN

May 2022 - Jun. 2024

- Developed modular, object-oriented Python software integrating machine learning (Jax) with physics-based models for data assimilation (ensemble Kalman filters), surrogate modeling, and reliability improvements.
- Collaborated with a multidisciplinary team to enhance reduced gas-surface chemistry models, applying machine learning-based uncertainty quantification and validation methods for tractable and accurate hypersonic flight simulations.
- Integrated physics-based models with data-driven components to fuse accelerometer sensor data, applying frequency-domain signal processing (FFT) to enable robust state estimation in dynamics systems.

University of Colorado

Boulder, CO

RESEARCH ASSISTANT

Aug. 2019 - May 2024

- Developed distribution-based uncertainty representations for real-time space weather state estimation, enabling probabilistic predictions from deterministic machine learning models in PyTorch.
- Collaborated with an interdisciplinary team to develop and deliver software using machine learning and MPI for parallel computation, which remains in operational use.
- Designed and optimized C++ stochastic enrichment operators to improve predictive accuracy of reduced-order physics-based models, deploying large-scale simulations on supercomputers using MPI, Slurm, and automated workflows.

University of Texas at Austin

Austin, TX

RESEARCH FELLOW

May 2016 - Aug. 2019

- Contributed to the open-source Transition State Atomic Simulation Environment (TSASE) Python library, enhancing global optimization methods for computational chemistry and materials applications. (<http://theory.cm.utexas.edu/tsase/>)
- Developed and maintained a database to facilitate collaborative sharing of simulation results.

Institute of Pure and Applied Mathematics

Los Angeles, CA

RESEARCH EXPERIENCE FOR UNDERGRADUATES PARTICIPANT

Jun. - Aug. 2018

- Collaborated with industry sponsors, a graduate mentor, and undergraduate students to apply data science and machine learning techniques in Python and MATLAB to simulate additive manufacturing processes.

Electric Reliability Council of Texas

CYBERSECURITY INTERN

Taylor, TX

May - Dec. 2017

- Improved the company's security posture through the creation of an automated Open-Source Intelligence program that alerts security analysts of threats to the company or its personnel on the Clearnet and dark web.
- Created educational phishing exercises to improve company-wide security awareness.

Awards & Distinctions

2025	Selected Participant , Rising Stars in Computational and Data Sciences	Austin, TX
2024	Recipient , Clive F. Baillie Memorial Scholarship	Trieste, Italy
2023	Recipient , Space Weather with Quantified Uncertainties Student Travel Fellowship	Cambridge, MA
	Winner of the Best Paper in Model Validation and Uncertainty Quantification , Conference Proceedings of the Society for Experimental Mechanics (SEM) Series	Austin, TX
2020	Recipient , Dean's Summer Research Fellowship	Boulder, CO
	Winner of the Student E-Poster Competition in the Technology, Engineering, and Math category , the American Association for the Advancement of Science (AAAS) annual meeting	Austin, TX
2018		

Journal and Conference Papers

7. **Rileigh Bandy**, Enrico Camporeale, Andong Hu, Thomas Berger, and Rebecca Morrison. Accurate and reliable uncertainty estimates for deterministic predictions: extensions to under and overpredictions. 2026. URL: <https://doi.org/10.48550/arXiv.2604.08755>
6. Rebekah White, **Bandy, Rileigh**, and Teresa Portone. Inference in the presence of model-form uncertainties: Leveraging a prediction-oriented approach to improve uncertainty characterization. 2026. URL: <https://doi.org/10.48550/arXiv.2601.04396>
5. **Rileigh Bandy**, Rebecca Morrison, Erin Mussoni, and Teresa Portone. Hybrid physics-data enrichments to represent uncertainty in reduced gas-surface chemistry models for hypersonic flight. 2025. URL: <https://doi.org/10.48550/arXiv.2509.08137>
4. **Rileigh Bandy**, Rachel Washington, Rebecca Morrison, and Teresa Portone. Isolating and quantifying uncertainties in the vibration isolation round-robin challenge. In *Proceedings of the IMAC XLIII Conference*, Orlando, FL, USA, January 2025
3. **Rileigh Bandy**, Teresa Portone, and Rebecca E. Morrison. Stochastic model correction for the adaptive vibration isolation round-robin challenge. In *Proceedings of the IMAC XLII Conference*, Orlando, FL, USA, January 2024. URL: https://doi.org/10.1007/978-3-031-68893-5_8
2. **Rileigh Bandy** and Rebecca Morrison. Stochastic model corrections for reduced Lotka-Volterra models exhibiting mutual, commensal, competitive, and predatory interactions. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(1), 2024. URL: <https://doi.org/10.1063/5.0159043>
1. **Rileigh Bandy** and Rebecca Morrison. Quantifying model form uncertainty in spring-mass-damper systems. In *Proceedings of the IMAC XLI Conference*, Austin, TX, USA, February 2023. Springer. URL: <https://doi.org/10.1007/978-3-031-37003-8>

Presented Work _____

- 13 conference presentations; 6 poster presentations

Service and Professional Activities _____

STUDENTS (Co-)MENTORED

- Paul Smith, M.S. (2025)
- Eric Pabon Cancel, Ph.D. student (2025)
- Rachel Washington, M.S. student (2024)
- Mayah Drayton, B.S. student (2024)

Graduate Mentor 2020 - 2022

ACCESS AND INCLUSION PEER MENTORING PROGRAM

- Served as a mentor for first-year undergraduate underrepresented minority students in Engineering.
- Met regularly with mentees to provide support as they transitioned to college.

Post Secondary Volunteer Tutor 2021

BOULDER “I HAVE A DREAM” FOUNDATION

- Tutored post-secondary students in STEM subjects.

INSTITUTIONAL SERVICE

- Co-Organizer, Sandia UQ Working Group

CONFERENCE MINISYMPOSIA (Co)-ORGANIZED

- SIAM Conference on Uncertainty Quantification (UQ26)
- SIAM Conference on Computational Science and Engineering (CSE25)
- 18th U.S. National Congress on Computational Mechanics (USNCCM18)

REVIEWER FOR JOURNALS

- Chaos: An Interdisciplinary Journal of Nonlinear Science
- Review of Scientific Instruments

Technical Skills _____

Programming Languages	Python, Julia, Bash, C++, Java, R, MATLAB, SQL, JavaScript, HTML.
Software Development	Git, Agile, Amazon Web Services, Docker.
High Performance Computing	MPI, OpenMPI, Slurm.